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Seminar: Trends in Machine Learning

Exposé for Trends in Machine Learning

Bias in Multilingual Foundation Models: A Layer-Wise Cross-Lingual Analysis

Motivation and Problem Statement

Multilingual foundation models such as mBERT and XLM-R represent a central trend in modern machine learning: large-scale pretraining across many languages using shared transformer architectures. While bias in English embeddings has been widely studied, much less is known about how multilingual contextual models encode bias across typologically diverse languages, especially in low-resource settings.

This project investigates gender-related bias in multilingual contextual representations across Arabic, Spanish, and Tigrigna. By reducing contextual embeddings to static embeddings via controlled aggregation, the study enables systematic bias quantification and layer-wise comparison under a unified protocol.

Objectives and Research Questions

The project aims to:

- Extract static embeddings from multilingual contextual models using a unified pipeline.
- Quantify gender bias across transformer layers.
- Compare bias patterns across mBERT, XLM-R, and a multilingual DistilBERT variant.
- Analyze agreement and disagreement between established bias metrics.

Research questions include:

- Do multilingual foundation models encode measurable gender bias consistently across languages?
- How does bias vary across transformer layers?
- Do different bias estimators yield stable conclusions?
- How do architectural trends (scaling vs. distillation) influence encoded bias?

Methodology

Models:

- bert-base-multilingual-cased
- xlm-roberta-base
- Multilingual DistilBERT variant

Shared Concept Inventory:

- 80 profession concepts (English pivot)
- 15 male and 15 female anchor terms per language
- Frequency filtering in each corpus

Steps of the Study

1. Collect corpora and validate concept frequency.
2. Extract contextual token representations for each target word.
3. Apply mean pooling over subword tokens.
4. Aggregate across $n = 20$ contexts to obtain static embeddings.
5. Compute bias scores at every transformer layer using three metrics.
6. Apply bootstrap sampling for confidence intervals.
7. Compare layer-wise trends across models and languages.

Expected Contribution

The project provides a strictly controlled cross-lingual analysis of bias in multilingual foundation models. It offers insights into how bias is distributed across layers, how architectural trends affect bias encoding, and how stable bias metrics are across typologically diverse languages.

Structure of the Paper

Introduction, Related Work, Methodology, Experimental Setup, Results, Cross-Lingual Comparison, Discussion, Conclusion.